LCD1602 + PCF8574 I²C Display Module

High-Precision Mechanical Dimensional Guide & Multi-View Engineering Specifications

HD44780 / AIP31066 | PCF8574T / PCF8574AT | 4-WIRE I²C INTERFACE | DUAL HEADER OPTIONS

Enclosure Design Critical Note: Standard LCD1602 modules with pre-soldered PCF8574 backpacks come with either **Straight** (perpendicular 90°) or **Bent / Right-Angle** (parallel) 4-pin I²C pin headers. Straight headers require at least **30.0 mm** total internal enclosure depth for standard DuPont wiring, whereas right-angle bent headers reduce required depth to **16.0 mm** but require **15.0 mm** lateral perimeter clearance for the cable plug.

FIGURE 1: FRONT VIEW (Face & Mounting Geometry) All dimensions in mm

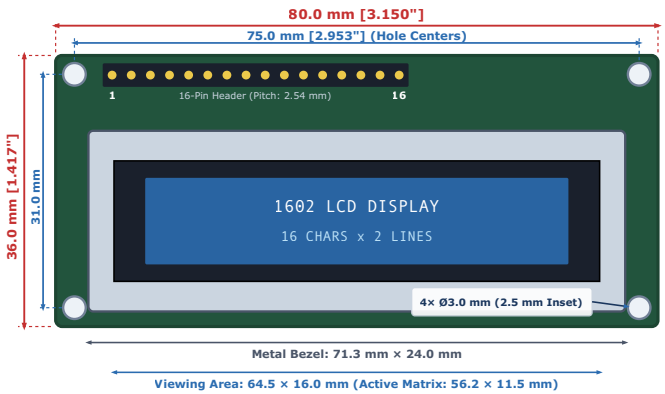


FIGURE 2: BACK VIEW (PCF8574 Backpack, Controls & Ports) Rear Hardware Components

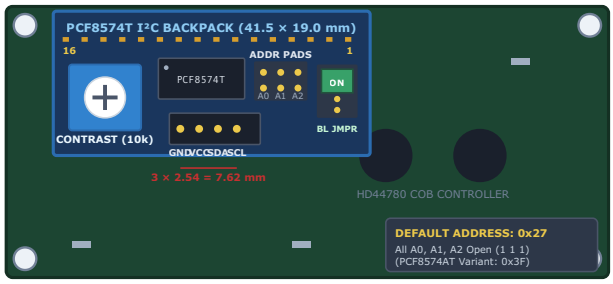


FIGURE 3: SIDE PROFILE & DEPTH STACKUP — STRAIGHT VS BENT (RIGHT-ANGLE) HEADERS Crucial Mechanical Clearance & Depth Comparison (Scale 5x)

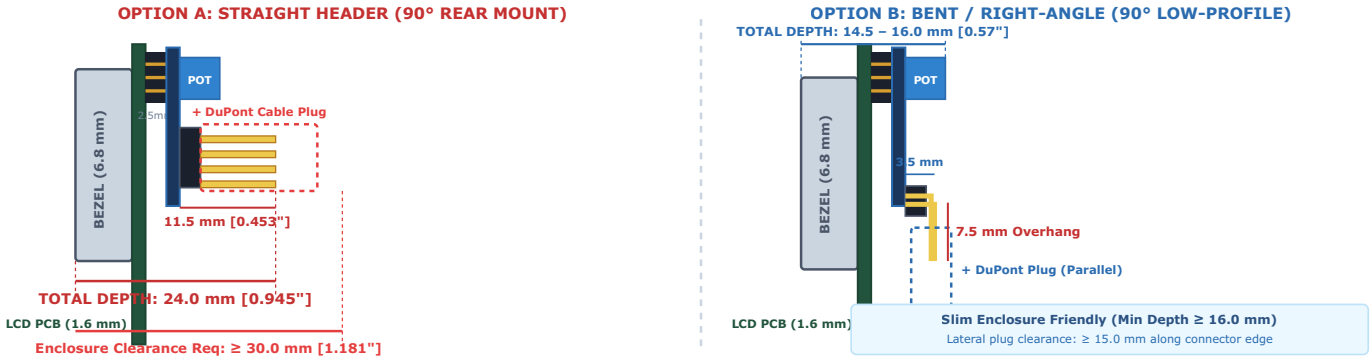
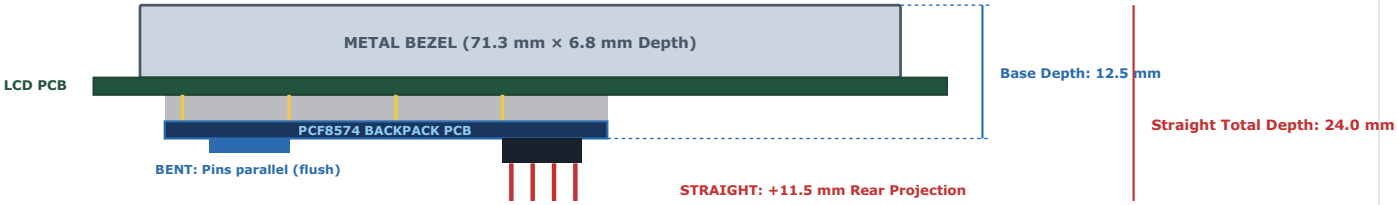


FIGURE 4: TOP PLAN VIEW (Overhead Depth & Header Pin Extension) Width: 80.0 mm | Bezel & Backpack Step-Down Top Profile



LCD1602 + PCF8574 I²C Electrical & Interface Reference

Pinout Assignments, Address Configuration, Electrical Specs & CAD Clearance

1. COMPREHENSIVE MECHANICAL SPECIFICATIONS & HEADER TOLERANCES

Parameter	Metric (mm)	Imperial (Inches)	Notes / Manufacturing Tolerance
Overall PCB Dimensions	**80.0 ± 0.3 × 36.0 ± 0.3**	3.150″ × 1.417″	FR4 Double-sided, 1.6 mm thickness
Mounting Hole Pitch (H × V)	**75.0 ± 0.2 × 31.0 ± 0.2**	2.953″ × 1.220″	Center-to-center; 2.5 mm inset from all edges
Mounting Hole Diameter	**Ø 3.0 mm** (or Ø 3.2 mm)	Ø 0.118″ – 0.126″	Accommodates standard M2.5 or M3 screws/ standoffs
Metal Bezel Outer Size	**71.3 × 24.0 × 6.8**	2.807″ × 0.945″ × 0.268″	Stainless steel / tin-plated bezel frame
Viewing Area (Window)	**64.5 × 16.0**	2.539″ × 0.630″	Minimum clear front panel cutout aperture
Active Display Matrix	**56.2 × 11.5**	2.213″ × 0.453″	16 characters × 2 rows, 5×8 dot matrix format
Character & Dot Size	2.95 × 4.35 mm (Dot: 0.55 × 0.65)	0.116″ × 0.171″	Pitch: 3.65 mm horizontal, 5.55 mm vertical
PCF8574 Backpack PCB Size	**41.5 ± 0.3 × 19.0 ± 0.3**	1.634″ × 0.748″	Soldered to 16-pin connector; 2.5 mm standoff gap
Depth: Straight Header Option	**24.0 mm max (Pin tips)**	**0.945″**	**Pin sticks straight out rear (11.5 mm from backpack)**
Enclosure Clearance (Straight)	**≥ 30.0 mm internal depth**	**≥ 1.181″**	**Required for DuPont female plug housing & wire relief**
Depth: Bent / Right-Angle Option	**14.5 – 16.0 mm max**	**0.571″ – 0.630″**	**Governed by trimpot height; pins bend parallel**
Enclosure Clearance (Bent)	**≥ 16.0 mm depth, ≥ 15 mm lateral**	**≥ 0.630″ depth, ≥ 0.59″ lat**	**Enables ultra-compact front-panel enclosures!**

2. I²C & 16-PIN INTERFACE PINOUT

Pin	Signal	Description / PCF8574 Mapping
1	**GND**	Common System Ground (0V)
2	**VCC**	+5.0V DC Power Supply (4.5V – 5.5V)
3	**SDA**	I²C Serial Data (Onboard 4.7kΩ pull-up to VCC)
4	**SCL**	I²C Serial Clock (Onboard 4.7kΩ pull-up to VCC)
LCD Interface Bus (Internal PCF8574 Connections)		
P0	RS	Register Select (0 = Command, 1 = Data)
P1	RW	Read/Write Select (Normally held LOW / 0 = Write)
P2	E	Enable Clock Strobe (Active High pulse)
P3	Backlight	Controls NPN transistor for LED Backlight
P4–P7	D4 – D7	High nibble 4-bit data bus to HD44780 controller

3. I²C ADDRESS SELECTION (PADS A0, A1, A2)

A2 Pad	A1 Pad	A0 Pad	PCF8574T (Default)	PCF8574AT Variant
OPEN (1)	**OPEN (1)**	**OPEN (1)**	**0x27 (Default)**	**0x3F (Default)**
OPEN (1)	OPEN (1)	SHORT (0)	0x26	0x3E
OPEN (1)	SHORT (0)	OPEN (1)	0x25	0x3D
OPEN (1)	SHORT (0)	SHORT (0)	0x24	0x3C
SHORT (0)	OPEN (1)	OPEN (1)	0x23	0x3B
SHORT (0)	OPEN (1)	SHORT (0)	0x22	0x3A
SHORT (0)	SHORT (0)	OPEN (1)	0x21	0x39
SHORT (0)	SHORT (0)	SHORT (0)	0x20	0x38

* Note: Bridge solder pads to GND to set to 0. Default open state uses onboard internal pull-ups (1 1 1).

4. ELECTRICAL CHARACTERISTICS

Parameter	Min	Typ	Max	Unit
Supply Voltage (V_{CC})	4.5	5.0	5.5	V
Operating Current (Backlight OFF)	1.2	2.0	3.0	mA
Backlight Current (LED ON)	15.0	22.0	30.0	mA
I²C Clock Frequency	-	100	400	kHz
Operating Temperature	-20	+25	+70	°C

5. HARDWARE CONTROLS & ADJUSTMENT GUIDE

Contrast Potentiometer (10 kΩ): If the display shows black boxes or is blank on first power-up, use a small flathead/Phillips screwdriver to rotate the blue potentiometer clockwise/counter-clockwise until character contrast is sharp.

Backlight Jumper Cap: Shunting the 2-pin jumper powers the LED backlight via the onboard transistor. Remove the jumper cap to fully disconnect backlight power or insert an external PWM signal for variable dimming.

6. OFFICIAL DOCUMENTATION & CONTROLLER LINKS

Component	Manufacturer / Specification	Official Datasheet / Documentation
HD44780U Controller	Hitachi / Renesas Dot Matrix LCD Controller	Hitachi HD44780U Official Datasheet (PDF)
PCF8574 / PCF8574A	Texas Instruments / NXP Remote 8-Bit I²C Expander	TI PCF8574 Remote I/O Expander Datasheet (PDF)
CircuitPython Library	Adafruit CircuitPython Character LCD	adafruit_character_lcd.character_lcd_i2c
Arduino Library	LiquidCrystal_I2C Library by Frank de Brabander	LiquidCrystal_I2C GitHub Repository